Modeling Customer Churn States as a Hidden Markov Model

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Abstract

Customer engagement naturally evolves over time, yet many churn and Customer Lifetime Value (CLV) models assume static buying behavior. This project applies a discrete Hidden Markov Model (HMM) to infer latent churn states using purchase frequency data from a major UK-based retailer. Transactions are transformed into a customer–month panel, aggregated into monthly purchase counts, and discretized using empirical quantiles to form emission categories. A three-state Categorical HMM is trained using expectation–maximization, revealing clear behavioral regimes corresponding to inactive, at-risk, and reactivation/active states. Transition and emission probabilities show structured patterns that align with customer engagement cycles, and Viterbi decoding yields interpretable most-likely state trajectories across customers. The results demonstrate that HMMs can recover meaningful churn pathways even with limited time horizons and simple count-based signals.

1 Problem Description

Understanding how customers transition between engagement levels is essential for predicting churn, revenue, and long-term CLV. Traditional CLV models frequently rely on independence assumptions or static behavioral parameters, making them ill-suited for capturing temporal engagement patterns. This project investigates whether a Hidden Markov Model can uncover latent churn dynamics using only monthly purchase counts.

The central question we address is: *Can latent engagement states be reliably inferred and interpreted from discrete transaction counts using an HMM framework?* This problem is fundamentally probabilistic: customer churn behavior is unobserved, but purchase outcomes provide indirect signals. HMMs incorporate temporal structure, latent states, and uncertainty, making them a natural modeling tool.

2 Data Sourcing and Processing

The dataset is sourced from the UCI Machine Learning Repository Online Retail Dataset. It contains all retail transactions from December 2010 through December 2011 for a UK gift retailer. Each invoice corresponds to a purchase, and each record contains item-level information including invoice ID, stock code, price, quantity, customer ID, and timestamp.

2.1 Preprocessing Steps

The dataset undergoes several standard transformations to construct a structured observation sequence:

1. **Datetime normalization.** The invoice timestamps are converted into month indices spanning `Month_0` through `Month_12`.
2. **Customer–month aggregation.** For each customer and each month, the total number of invoices is counted:

$$\text{Count}_{i,t} = |\{\text{purchases for customer } i \text{ in month } t\}|.$$

3. **Wide-format customer matrix.** A pivot table is created so that each row represents a customer and each column represents a month:

$$X_i = [x_{i,0}, x_{i,1}, \ldots, x_{i,12}].$$

   Missing months are filled with zeros.
4. **Quantile-based discretization.** Nonzero monthly purchase counts are used to estimate empirical quartiles. The 25th and 75th percentiles correspond to approximately 1–9 and 10–35 purchases, respectively. We define emission categories:

$$0:0, \quad 1:1-9, \quad 2:10-35, \quad 3:35+.$$

5. **Equal-width bin discretization.** As another method, we take the min and max of non-zero purchase counts and evenly divide the interval into 3 boundaries (4 bins).
6. **Final observation sequence format.** All customer sequences are flattened into a single list of categorical emission values, one for each month–customer pair. All sequences are of equal length (13 months), enabling batch training.

These preprocessing steps standardize the data into a form suitable for categorical HMM training.

3 Modeling and Inference

We adopt a discrete-time Hidden Markov Model with categorical emissions, which assumes that purchase behavior is governed by an underlying unobserved engagement state that evolves according to a Markov chain.

3.1 Model Structure

Let:

$$Z_t \in \{0, 1...K\} \quad \text{(hidden states)}$$

$$X_t \in \{0, 1, 2, 3\} \quad \text{(emission categories)}$$

The model contains:

$$\pi = \text{initial state distribution}$$

$$A = \text{state transition matrix}$$

$$B = \text{emission probabilities}$$

The hidden state dynamics follow:

$$P(Z_{t+1} = j \mid Z_t = i) = A_{ij}$$

and emissions follow:

$$P(X_t = k \mid Z_t = i) = B_{ik}.$$

3.2 Parameter Learning

Parameters are learned using the expectation–maximization algorithm, which iteratively maximizes the expected complete-data log-likelihood:

$$\theta^{(m+1)} = \arg\max_{\theta} \mathbb{E}\left[\log p(X, Z \mid \theta) \mid X, \theta^{(m)}\right].$$

The model is trained on all customers simultaneously, with each customer represented as a length-13 categorical emission sequence. A random seed is set for reproducibility.

To evaluate the robustness of the Hidden Markov Model, we conducted a sensitivity analysis by varying two key modeling components: (1) the number of hidden states ($K = 2, 3, 4$) and (2) the emission binning method, comparing percentile-based cutoffs against equal-width discretization of monthly purchase counts. For each configuration, we re-estimated the HMM using the EM algorithm and recorded the final log-likelihood value, which measures how well the model explains the observed customer purchase sequences. Higher (less negative) log-likelihood values shows better fit.

3.3 State Decoding

The Viterbi algorithm is used to compute the most probable sequence of hidden states:

$$Z^* = \arg\max_{Z} p(Z, X \mid \theta).$$

This yields interpretable monthly engagement trajectories for each customer.

4 Results and Discussion

4.1 HMM Sensitivity Analysis

Hidden States	Binning Strategy	Log-Likelihood
2	Percentile	$-42{,}881$
2	Equal-Width	$-28{,}892$
3	Percentile	$-42{,}871$
3	Equal-Width	$-28{,}887$
4	Percentile	$-42{,}815$
4	Equal-Width	$-31{,}151$

Table 1: Log-likelihood comparison for HMM sensitivity analysis.

The results show a consistent pattern. First, equal-width binning outperforms percentile-based binning across all values of K. This shows that percentile thresholds may group too many observations into the same categories, while equal-width bins balances it out. Second, models with $K = 3$ and $K = 2$ under equal-width binning achieve the highest log-likelihood values. This shows that the binning method affects the model's performance more than the number of hidden states that is tested.

From a business perspective, this means that customer purchasing behavior is better categorized when purchase activity is grouped into fixed ranges rather than relative percentile thresholds.

The best model seems to be the one with equal-width bins and 3 states. Further analysis in our paper will use this model.

4.2 Model Convergence and Stability

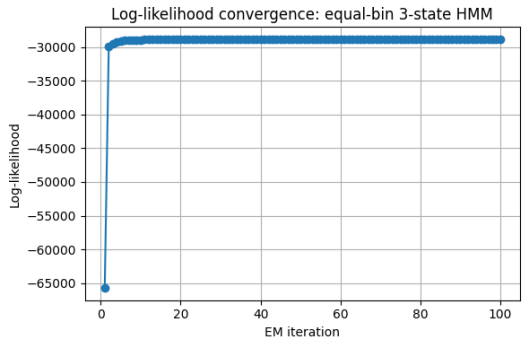

Figure 1: Log-Likelihood per iteration of EM

The expectation–maximization procedure exhibits monotonic log-likelihood improvement across iterations and converges within the set number of iterations. This indicates that the model reaches a stable set of state and emission probabilities under the chosen discretization and structure.

More data would likely help. The current dataset only records observations over a 13-month period, which might not be sufficient to capture lifetime customer relations. We discuss this more on the subsection on Limitations.

4.3 Initial State Distribution

The estimated initial probabilities are as follows.

Table 2: Initial state distribution for the best equal-bin 3-state HMM.

	Initial Probability
State_0	0.8477
State_1	0.1079
State_2	0.0445

We can see at the beginning, most customers start in State_0. Note that this reflects the customer distribution at Month_0, not necessarily their "true" initial state when they first became customers.

4.4 State Transition Dynamics

Table 3: State transition matrix for the best equal-bin 3-state HMM.

	State_0	State_1	State_2
State_0	0.1656	0.0003	0.8341
State_1	0.0189	0.9695	0.0116
State_2	0.4572	0.0224	0.5203

4.5 Emission Probabilities

Table 4: Emission probabilities (Zero, Low, Medium, High) for the best equal-bin 3-state HMM.

	Zero	Low	Medium	High
State_0	0.8405	0.1595	0.0000	0.0000
State_1	0.2837	0.7145	0.0013	0.0005
State_2	0.8463	0.1537	0.0000	0.0000

State 0 and State 2 exhibit the highest probability of Zero purchases.

4.6 State Interpreations

- **State 0: Churn/Inactivity**. Because the initial probability of 0.85 and emission probability P(Zero | State 0) = 0.84 show most customers start here and rarely transact. Transition matrix indicates a 0.83 chance of leaping directly to State 2 and almost zero flow to State 1, so the few reactivations skip the "low" state altogether. Such behavior matches dormant accounts that occasionally return with a spike. They spend most months with zero purchases and only reactivate in bursts.
- **State 1: Low purchasers**. Because emission 'P(Low | State 1) = 0.71' (with only 0.28 zero) and an extremely high self-transition 'P(State 1 $\rightarrow$ State 1) = 0.97' reflect consistent, low-volume activity. Customers in this state keep buying small amounts month after month, rarely churning (only 2% move to State 0) and rarely upgrading to the bursty State 2 (1%).
- **State 2: Reactivated**. Half of the customers in this state churn right back to State 0 (0.46), only 2% drift into State 1, and the rest (0.52) linger briefly. Emission probabilities still favor zero purchases (0.85 zero vs 0.15 low) because equal-width bins make the high bin unused, meaning the state captures the immediate aftermath of a reactivation rather than sustained high spend. This "burst" state probably represents customers who just returned after inactivity—they spike out of State 0.

4.7 Viterbi State Sequences

Running the Viterbi Algorithm allows us to find the most likely sequence of hidden states for each customer. We ran the viterbi algorithm on the first 5 customers:

Table 5: Viterbi States for First Five Customers

CustomerID	Month_0	Month_1	Month_2	Month_3	Month_4	Month_5	Month_6	Month_7	Month_8	Month_9	Month_10	Month_11	Month_12
12346.000000	0	2	0	2	2	0	2	0	2	0	2	0	2
12347.000000	1	1	1	1	1	1	1	1	1	1	1	1	1
12348.000000	0	2	0	2	0	2	0	2	2	0	2	0	2
12349.000000	0	2	0	2	0	2	2	0	2	0	2	0	2
12350.000000	0	2	0	2	0	2	0	2	0	2	0	2	2

The decoded sequences reveal some behavioral structure:

- Most customers spend a majority of months in State 2 or State 0, meaning most customers are inactive with only short bursts of activity.
- State 1 is stable (high probability of staying at State 1), representing a customer with consistent activity and purchases.

These patterns match typical ordering cycles for wholesale or bulk-purchase businesses.

4.8 Qualitative Insights

The HMM successfully captures the following:

- Customers fluctuate between prolonged inactivity and occasional bursts.

- Reactivation tends to occur in short windows.
- Consistent high purchases are rare. Most active customers only buy small amounts regularly.

4.9 Limitations

The main limitation is from the limited period of time recorded in the dataset. Because of that, the initial state probabilities reflect the distribution at Month_0, not the true initial state when customers first joined. This means the initial state probabilities might not generalize to completely new customers and their lifetime interaction with this company.

The transition probabilities for Inactive to Active or semi-Active might also be inflated due to the nature of our dataset, which is filled with customers who eventually make a purchase in that period. While in reality, there could be customers who were active outside of the time period and are inactive during the time period. These customers are not captured in the dataset at all.

5 Conclusion

This project demonstrates that a discrete Hidden Markov Model can recover meaningful latent churn and engagement states using only monthly purchase counts. Emission and transition patterns strongly differentiate inactive, at-risk, and active customer behaviors. The model converges reliably, produces interpretable CPTs, and yields plausible state trajectories.

The model could be improved by observing a longer time period, using customer attributes, and incorporating purchase value.

6 Reflections & Contributions

6.1 Suggestions for future students

- Start early.
- Try to find a topic that's useful and you find interesting. It makes doing the project more fun.
- Once you decide on a topic, you should think carefully how you can model it as a Bayesian Network, and then think about what can you do with it that would yield useful results.

6.2 Team Contributions and Reflection

Each member contributed to report writing.

- **Daniel**: Model construction and running the Viterbi algorithm.
  Reflection: I quite liked this project. Homeworks and exams so far had us solving pre-packaged problems with Bayesian Networks, but in this project I had to think about what HMMs could be used for and how it could be useful/actionable. I now realize there are quite a lot of real-life applications for Bayesian Networks.
- **Flavia**: Data preprocessing (EDA and data cleaning for HMM use)
  Reflection: It's great to get to put theory into practice and getting a glimpse into where the theories that we learn in class also fits into the bigger data science pipeline. The project made the topics that we learn in class to connect and build upon each other to reach the objective that we set with this project and find insights that can be useful to make decisions moving forward.
- **Varick**: Data preprocessing (EDA and data cleaning for HMM use)
  Reflection: I liked this project a lot. My favorite part was brainstorming what real-world scenarios we could apply HMMs too, and me and the group were discussing how businesses plan customer retainment. When we researched further, we found out about Customer Lifetime Value, and it was exciting that we could model it with an HMM. When we cleaned the data, modeled it, and actually found that it made sense for an HMM, I became more exctied for the project.

- **Calvin**: Model construction.
  Reflection: This project taught me how customer behavior changes over time and how a Hidden Markov Model can help us understand those changes. I learned that even simple monthly purchase data can reveal clear patterns. Overall, it showed me how data and modeling can explain real-life decisions.
- **Tristan**: Model construction.
  Reflection: I really enjoyed this project because it pushed me to think more deeply about how Hidden Markov Models actually work in practice. Most of the time in class, we apply HMM formulas to small, defined problems, but building the model from scratch helped me see how each component fits together to represent real behavior. Working on model construction made me pay attention to the assumptions we were making and how those assumptions affected the results. It was interesting to see how a simple structure could still capture meaningful patterns in the data.

Gen AI tools were used only to brainstorm high level ideas and verify library usage details.

References

UCI Machine Learning Repository: Online Retail Dataset.
`https://archive.ics.uci.edu/dataset/352/online%2Bretail`